

OOP with Java - 25.3

In-Class Test - 02

Time: 30Mins

Answer ALL questions.

Write your answers clearly and legibly. Marks are awarded for correct logic, syntax, and structure.

No electronic devices are permitted.

This is a closed-book examination.

(100marks)

-
1. Create a class called **Vehicle** with suitable access modifiers and data types to store a **vehicle ID, brand, speed, and whether it is electric**. Write a parameterized constructor and getters/setters for all variables.
 - a. Create a **subclass** called **Car** that have Is-a relationship with **Vehicle**. Add a suitable access modifiers and data types to store **numDoors**. Write a parameterized constructor.
 - b. Write a method called **displayInfo()** in Vehicle class that prints all vehicle details. Perform displayInfo() in Car class and also print numDoors.
 - c. In **main()**, create a Vehicle object and a Car object. Call displayInfo() on both and print the results.

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 - c. In **main()**, create a Vehicle object and a Car object. Call displayInfo() on both and print the results..

2. Create a class called **Person** with suitable access modifiers and data types to store **person ID, name, age, and email**. Write a parameterized constructor and getters/setters for all variables.
 - a. Create a subclass called **Student** that have relationship to **Person**. Add suitable access modifier and data type **gpa**. Write a parameterized constructor.
 - b. Write a method called **displayDetails()** in **Person** class that prints all details. Perform **displayDetails()** in Student class to also print **gpa**.
 - c. In **main()**, create a Person object and a Student object. Call displayDetails() on both and print the results.

4. Create a class called **Person** with suitable access modifiers and data types to store **person ID, name, age, and email**. Write a parameterized constructor and getters/setters for all variables.
 - d. Create a subclass called **Student** that have relationship to **Person**. Add suitable access modifier and data type **gpa**. Write a parameterized constructor.
 - e. Write a method called **displayDetails()** in **Person** class that prints all details. Perform **displayDetails()** in Student class to also print **gpa**.
 - f. In **main()**, create a Person object and a Student object. Call displayDetails() on both and print the results.